A Unified Homotopy Framework for Nonlinear System Identification and Simulation: Constructive Dimensioning of Bilinear Neural Architectures

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Abstract

We develop a unified algorithmic framework for nonlinear system identification and forward simulation of ordinary differential equations, built on Liao's zeroth-order homotopy deformation equation. Both computational tasks—fitting a parametric function approximator from input–output data, and forward-integrating the resulting nonlinear ODE—are reduced to closed-form Newton (z_1) and Halley (z_2) corrections derived from the same deformation equation. The framework exploits the bilinear parameter structure of common function representations (radial basis functions, sigmoid multilayer perceptrons): a linear last-layer that admits an exact one-step least-squares solution, and nonlinear internal parameters that admit curvature-aware corrections without a scalar learning rate.

The principal theoretical contribution is a constructive dimensioning chain $\varepsilon_{\mathrm{sim}} \to \varepsilon_{\mathrm{id}} \to M_{\min} \to N_{\min} \to T_{\max}$ that maps a target forward-simulation accuracy to the minimum number of basis functions, the minimum number of training samples, and the maximum admissible integration time step. The chain is constructive in the sense that each link is computable from approximation-theoretic bounds and system quantities. An operator-theoretic interpretation emerges: the choice of the auxiliary linear operator $\mathcal{L}$ encodes prior knowledge that reduces the residual nonlinearity the approximator must

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learn, and therefore reduces the required approximator dimension.

We validate the framework on benchmark problems: a one-dimensional discharge-flow system (orifice-tank model), a two-dimensional Lotka-Volterra predator-prey system, and robustness experiments under measurement noise. Gaussian RBF and sigmoid MLP approximators identified from sparse data (20–40 points) attain identification residuals from $3 \cdot 10^{-2}$ (RBF) down to machine precision ($1.7 \cdot 10^{-8}$, MLP) in a single outer iteration and yield forward-simulation errors of at most 2.3% over 200-step horizons on the 1D benchmark. The scope of the empirical validation is deliberately narrow—low-dimensional ODEs, small-scale approximators—and the role of the present work is to establish the algorithmic framework, the dimensioning theory, and the proof-of-concept computational regime; broader empirical assessment and extension to high-dimensional or stiff systems are identified as follow-up work.

Keywords: Homotopy analysis method, nonlinear system identification, ordinary differential equations, radial basis functions, Newton–Halley corrections, constructive dimensioning, bilinear approximation, function approximation theory, computational methods

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1. Introduction

Consider a nonlinear dynamical system observed through input–output data,

$$\dot{y}(t) + f(y(t)) = u(t), \qquad y(0) = y_0, \tag{1}$$

where f is an unknown nonlinearity and $u(t)$ is a measured excitation. Two classical computational problems arise: *identification*—reconstruct f from data $\{(t_i, y_i, u_i)\}_{i=1}^{N}$ using a parametric function approximator—and *forward simulation*—given an identified $\hat{f}$, integrate (1) forward for new inputs and initial conditions. The two are usually attacked with separate computational machinery. Identification employs gradient-based optimization of parametric approximators (radial basis functions, multilayer perceptrons [20, 35, 43, 34]) or, more recently, sparse symbolic regression [9, 13]. Simulation employs explicit and implicit time-stepping schemes [19, 11]. Each approach has its own convergence theory, tuning parameters, and failure modes.

This paper isolates an algebraic structure that is common to both problems and exploits it for joint, hyperparameter-light treatment. The starting point is the zeroth-order deformation equation of the Homotopy Analysis

Method [26, 28, 27],

$$(1-q)\,\mathcal{L}[\phi(q)-\phi_0] = q\,c_0\,\mathcal{N}[\phi(q)], \qquad q \in [0,1], \tag{2}$$

in which $\mathcal{L}$ is a chosen linear operator, ϕ_0 is an initial approximation, and c_0 is a convergence-control parameter. Equation (2) is agnostic to the nature of ϕ and $\mathcal{N}$: ϕ may be a parameter vector and $\mathcal{N}$ an approximation residual, or ϕ may be a state trajectory and $\mathcal{N}$ a discretised ODE residual. Truncating the homotopy series and matching powers of q yields three closed-form corrections z_1, z_2, z_3 whose algebraic shape depends only on the local Taylor coefficients of $\mathcal{N}$ at ϕ_0, not on the dimension or interpretation of ϕ.

Contributions. The main contributions are the following.

(C1) *Universal deformation algebra.* We show that the corrections z_1 and z_2, when read off the deformation equation (2), recover Newton's method (with the choice $\lambda = \mathcal{N}_0'$, $\sigma = -1$, $z_2 = 0$) and Halley's method as instances, and admit a parametric extension to arbitrary λ. This places HAM, Newton, and Halley in a common scalar parametrisation (γ, σ) that is exploited uniformly across identification, simulation, and the linear/nonlinear sub-problems within identification.

(C2) *Bilinear factorisation of identification.* For both RBF and MLP approximators, the identification residual is linear in the last-layer weights and nonlinear in the internal parameters. The framework solves the linear sub-problem exactly in one step (z_1 with quadratic form, $z_2 = z_3 = 0$ structurally) and applies homotopy corrections only to the nonlinear sub-problem. This is a Variable-Projection-style separation [17, 44] but with curvature-aware corrections instead of gradient descent on the projected residual.

(C3) *Derivative-free integral simulation.* For the simulation step we recast (1) as a Volterra integral equation with exponential kernel $K(t,s) = e^{-k(t-s)}$, where k is the linearisation of the identified nonlinearity at the operating point. The discrete residual then has all coefficients of order $O(1)$ and contains no $1/T$ factor; sensor noise is attenuated rather than amplified. The homotopy corrections on this residual give per-step convergence in 1–2 evaluations.

(C4) *Constructive dimensioning.* We give explicit conditions linking target simulation error ε_{sim} to identification error ε_{id} (Gronwall propagation), to the minimum number of basis functions $M_{\min}$ (RBF approximation rate à la Buhmann 2003; Barron norm 1993 for sigmoid MLPs), to the minimum number of training samples $N_{\min}$ (Jacobian conditioning), and to the maximum admissible step $T_{\max}$ (Newton convergence on the integral residual). The chain is constructive in the sense that each link is computable from system quantities and the others are determined.

(C5) *Inductive bias as operator choice.* The dimensioning chain shows that $M_{\min}$ depends not on $\|f''\|_\infty$ globally but on $\|\tilde{f}''\|_\infty$, where $\tilde{f} = f - \mathcal{L}^{-1}[f]$ is the residual nonlinearity left after the linear operator absorbs the part of f that is already known. Better prior knowledge produces a smaller $\tilde{f}$ and reduces the network size needed for a given accuracy. This formalises a quantitative meaning of "inductive bias" that is operator-side rather than loss-side, in contrast to Physics-Informed Neural Networks (PINNs) [38] where physics enters the loss as a soft penalty.

Scope of the empirical study. The numerical results in Section 9 cover a one-dimensional first-order ODE (orifice-tank discharge) with two architectures (5-centre RBF, 8-unit sigmoid MLP) and 20 training points. The role of these experiments is to verify, in the simplest non-trivial setting, that the closed-form corrections converge as predicted, that the linear/nonlinear separation behaves as analysed, and that the dimensioning chain is consistent with the observed network sizes. Generalisation to multi-dimensional, stiff, or chaotic dynamics, and to convolutional, recurrent, or attention-based architectures, is consistent with the bilinear structure but requires follow-up work; we discuss this explicitly in Section 10.

Outline. Section 2 positions the contribution against gradient-based training, Newton-type methods for neural networks, PINNs and Neural ODEs, and symbolic regression. Section 3 states the deformation equation and derives the universal scalar corrections. Section 4 develops the identification level for RBF and MLP architectures. Section 5 develops the integral simulation level. Section 6 states the unification result in compact form. Section 7 establishes the constructive dimensioning chain. Section 8 interprets the chain as a formalisation of inductive bias. Section 9 reports the numerical demonstrations. Section 10 delineates the empirical scope and identifies follow-up work. Section 11 clarifies the relationship to companion manuscripts by the authors. Section 13 concludes.

2. Related work

Gradient-based neural network training. Backpropagation [42] computes the gradient of a scalar loss with respect to all parameters in a single backward pass; first-order methods such as stochastic gradient descent and Adam [1] then update the parameters via a learning-rate-scaled step in the negative-gradient direction. The combination is the workhorse of modern deep learning, but for the small networks and physically-interpretable problems we consider here it has well-known disadvantages: a learning rate must be tuned, convergence is at best linear in the loss, and the quality of the final iterate depends on initialisation and stopping.

Newton-type and second-order methods for neural networks. Second-order methods exploit curvature information through the Hessian or an approximation thereof. Becker and LeCun [5] proposed a diagonal Hessian update for MLPs. Martens [31] introduced Hessian-free optimisation, which builds Newton steps via conjugate gradient on Hessian-vector products. The K-FAC family [32] approximates the Hessian by a Kronecker-factored block-diagonal structure. The framework of the present paper is closer in spirit to Sjölund et al. [44], who exploit the linear/nonlinear separation of single-hidden-layer networks via Variable Projection. The principal differences are: (i) we derive the corrections z_1, z_2 from the deformation equation rather than from a Levenberg–Marquardt or Gauss–Newton starting point, which gives a uniform parametrisation across identification and simulation; (ii) we apply the same corrections to ODE simulation, not only to identification; and (iii) we add an explicit dimensioning analysis that links the corrections' convergence conditions to network size and data sufficiency.

Physics-Informed Neural Networks. PINNs [38, 23] embed the governing differential equation into a soft penalty in the training loss and minimise the combined data-plus-physics loss with backpropagation. The architecture is left untouched. Two consequences follow: (i) the network size must be chosen heuristically because the physics term does not change the function class, only the search direction; (ii) the loss landscape becomes ill-conditioned when the physics weight is large or the system is stiff [47, 25]. The framework of the present paper places physics in the auxiliary linear operator $\mathcal{L}$ and consequently in the residual nonlinearity $\tilde{f} = f - \mathcal{L}^{-1}[f]$ that the network must approximate. The effect on architecture is constructive: better prior knowledge shrinks $\|\tilde{f}''\|_\infty$ and therefore the required M.

Neural Ordinary Differential Equations. Chen et al. [15] parametrise the right-hand side of an ODE by a neural network and train it via the adjoint method. This is conceptually orthogonal to our work: their primary problem is to back-propagate through the ODE solver efficiently for general loss functions on the trajectory; our primary problem is to identify f from sparse data and then simulate accurately. Rackauckas et al. [37] unify these directions under the Universal Differential Equations programme, where physics-known terms are combined with neural-network-learned residual terms; that combination is precisely the operator-side decomposition $f = \mathcal{L}^{-1}[\cdot] + \tilde{f}$ that we use here, and our dimensioning analysis applies to it.

Symbolic and sparse identification. SINDy [9] and its extensions [13, 22] reconstruct f from data by sparse regression on a candidate library of nonlinear basis functions, with ℓ_1-style penalties to enforce parsimony. The output is a symbolic expression rather than a black-box approximator. Our framework is intermediate between SINDy and classical neural identification: the basis is a smooth parametrised family (RBF, MLP) rather than a fixed library, but the linear sub-problem is solved exactly as in SINDy and the nonlinear corrections are closed-form rather than gradient-based.

Reservoir computing and echo-state networks. Lukoševičius and Jaeger [29] train only the linear output layer of a recurrent network, freezing the nonlinear hidden dynamics. The linear/nonlinear split is identical in spirit to the one we use, restricted to the case in which the nonlinear part is fixed by design. Our framework treats both halves: the linear part exactly, the nonlinear part by curvature-aware corrections.

Homotopy methods in numerical analysis and control. The Homotopy Analysis Method [26, 28] was developed for analytic series solutions to nonlinear differential and integral equations. Turkyilmazoglu [45] and Vajravelu and Van Gorder [46] analysed convergence properties and the role of the auxiliary parameter c_0. Within optimisation, Allgower and Georg [2] reviewed numerical continuation methods, of which homotopy is a special case. The contribution of the present paper is to use the deformation equation as an *algorithmic* primitive—truncated, evaluated numerically, and applied uniformly to identification and simulation—rather than as a device for analytic series construction.

3. The universal deformation equation

3.1. Liao's zeroth-order deformation

Given a nonlinear operator $\mathcal{N}$ acting on an unknown ϕ with $\mathcal{N}[\phi^*] = 0$, the deformation equation (2) constructs a one-parameter family $\phi(q)$, $q \in [0, 1]$, such that $\phi(0) = \phi_0$ and $\phi(1) = \phi^*$. The freedom in choosing $\mathcal{L}$, ϕ_0, and c_0 is the analytic content of the method [27]. We use this freedom algorithmically: $\mathcal{L}$ encodes prior knowledge, ϕ_0 is the best available cheap approximation, and c_0 scales the corrections.

3.2. The universal scalar corrections

Expanding $\phi(q) = \phi_0 + \sum_{m\geq 1} z_m\, q^m$ and Taylor-expanding $\mathcal{N}[\phi_0 + \delta]$ about ϕ_0, matching coefficients of q^m in (2) yields, in scalar form,

$$\boxed{z_1 = \gamma, \qquad z_2 = \gamma(1+\sigma), \qquad z_3 = \gamma(1+\sigma)^2 + \frac{c_0\,\mathcal{N}_0''}{2\,\lambda}\,\gamma^2,} \tag{3}$$

where

$$\gamma \triangleq \frac{c_0\,\mathcal{N}_0}{\lambda}, \qquad \sigma \triangleq \frac{c_0\,\mathcal{N}_0'}{\lambda}, \tag{4}$$

$\mathcal{N}_0 = \mathcal{N}[\phi_0]$, $\mathcal{N}_0' = \partial\mathcal{N}/\partial\phi\big|_{\phi_0}$, $\mathcal{N}_0'' = \partial^2\mathcal{N}/\partial\phi^2\big|_{\phi_0}$, and λ is the scalar representation of $\mathcal{L}$. For finite-dimensional problems λ is a chosen positive scalar; for differential operators λ incorporates the discretisation. The truncated series converges if $|1+\sigma| < 1$, equivalently $0 < \mathcal{N}_0'/\lambda < 2$ for $c_0 = -1$.

Remark 1 (Newton and Halley as special cases)**.** *With $\lambda = \mathcal{N}_0'$ and $c_0 = -1$ one obtains $\sigma = -1$, hence $z_2 = 0$ and z_3 reduces to the classical Halley correction $-\frac{1}{2}\mathcal{N}_0^2\,\mathcal{N}_0''/(\mathcal{N}_0')^3$ [18]. The expressions (3) therefore generalise Newton–Halley to arbitrary λ, with the σ-parametrised one-parameter family interpolating between the two and beyond.*

Remark 2 (Vectorial reading)**.** *For $\phi \in \mathbb{R}^n$, the scalar $\mathcal{N}_0'$ is replaced by a Jacobian and $\mathcal{N}_0''$ by a Hessian tensor. With $\lambda = J^T J$ (Gauss–Newton metric) and a diagonal Hessian approximation, the expressions (3) produce the Levenberg–Marquardt step at order one and a Halley-type correction at order two. The diagonal Hessian approximation is a deliberate choice to retain $O(NM)$ cost per step; we discuss its accuracy in Section 4.*

4. Identification by homotopy

4.1. Problem statement and bilinear factorisation

We represent the unknown nonlinearity by a parametric approximator with linear last layer,

$$\hat{f}(y) = \mathbf{\Phi}(y; \boldsymbol{\eta})\, \mathbf{w}, \tag{5}$$

where $\mathbf{\Phi} : \mathbb{R} \to \mathbb{R}^{1\times M}$ is a row of basis functions parametrised by $\boldsymbol{\eta}$ (centres and widths for an RBF; hidden weights and biases for an MLP), and $\mathbf{w} \in \mathbb{R}^M$ are linear weights. Given training data $\{(y_i, f_i)\}_{i=1}^N$, the identification residual is

$$\mathcal{N}_{\text{id}}(\boldsymbol{\eta}, \mathbf{w}) = \mathbf{\Phi}(\boldsymbol{\eta})\, \mathbf{w} - \mathbf{f}, \tag{6}$$

where $\mathbf{\Phi}_{ij} = \mathbf{\Phi}(y_i; \boldsymbol{\eta})_j$ is the $N \times M$ design matrix and $\mathbf{f} = (f_1, \ldots, f_N)^T$.

The residual is *linear* in $\mathbf{w}$ and *nonlinear* in $\boldsymbol{\eta}$. We exploit this by alternating an exact linear solve on $\mathbf{w}$ with curvature-aware corrections on $\boldsymbol{\eta}$.

4.2. Linear sub-problem (last-layer weights)

For fixed $\boldsymbol{\eta}$, the optimal weights are

$$\mathbf{w}^* = (\mathbf{\Phi}^T\mathbf{\Phi})^{-1}\mathbf{\Phi}^T\mathbf{f}. \tag{7}$$

This is the deformation correction z_1 with $\mathcal{L} = \mathbf{\Phi}^T\mathbf{\Phi}$ applied to the linear residual; structurally, $\partial^2\mathcal{N}_{\text{id}}/\partial\mathbf{w}^2 = 0$, so $z_2 = z_3 = 0$ and the solution is exact in one step.

4.3. Nonlinear sub-problem (internal parameters)

After substituting $\mathbf{w} = \mathbf{w}^*(\boldsymbol{\eta})$, the residual is a function of $\boldsymbol{\eta}$ only. The Jacobian J_η is computed analytically; for an RBF with centre c_j and width λ_j,

$$\frac{\partial \mathcal{N}_i}{\partial c_j} = w_j\, \varphi'\!\left(\frac{|y_i - c_j|}{\lambda_j}\right) \frac{\text{sgn}(y_i - c_j)}{\lambda_j}, \tag{8}$$

$$\frac{\partial \mathcal{N}_i}{\partial \lambda_j} = -w_j\, \varphi'\!\left(\frac{|y_i - c_j|}{\lambda_j}\right) \frac{|y_i - c_j|}{\lambda_j^2}. \tag{9}$$

For an MLP the corresponding derivatives are obtained by analytical differentiation of the activation, e.g. $\sigma'(x) = \sigma(x)(1 - \sigma(x))$ and $\sigma''(x) = \sigma'(x)(1-2\sigma(x))$ for the logistic sigmoid. We emphasise that these closed-form derivatives *are* the parameter-by-parameter chain rule and so are mathematically equivalent to backpropagation; the difference is downstream of the gradient. We do not use a learning rate or stochastic batches.

The Newton correction on the nonlinear parameters is

$$\Delta\boldsymbol{\eta}_1 = -(J_\eta^T J_\eta)^{-1}\, J_\eta^T\, \mathcal{N}_0, \tag{10}$$

which is z_1 of (3) read in vectorial form with $\lambda = J_\eta^T J_\eta$. For the Halley correction we use a *diagonal* Hessian approximation,

$$\Delta\eta_{j,2} \approx -\tfrac{1}{2}\, \frac{(\mathcal{N}_1^T\, \partial^2\mathcal{N}/\partial\eta_j^2)\,(\Delta\eta_{j,1})^2}{(J_{\eta j}^T J_{\eta j})}, \tag{11}$$

where $\mathcal{N}_1$ is the residual after the Newton step.

Diagonal-Hessian approximation: scope and effect. Equation (11) retains only the j-diagonal entries of the parameter Hessian. The exact Halley step contains additional off-diagonal contributions that scale with the off-diagonal Hessian entries. The diagonal approximation is exact when the Hessian is diagonally dominant (in particular when the basis functions φ_j have non-overlapping supports relative to the data) and is otherwise a controlled approximation that preserves quadratic convergence under the same conditions as Newton. We use it because (i) it keeps the per-iteration cost at $O(NM)$ and (ii) it is the form that survives the universal scalar parametrisation (3). For problems in which the basis functions overlap strongly, a Kronecker-factored Hessian [32] can replace the diagonal in (11) without changing the rest of the framework.

4.4. Algorithm

Algorithm 1 states the identification procedure.

5. Simulation by integral homotopy

5.1. Operator splitting and analytic starting solution

After identification, the simulation problem is to integrate $\dot{y} + \hat{f}(y) = u(t)$ from $y(0) = y_0$. We linearise the identified RHS at a reference point $\bar{y}$,

$$k = \hat{f}'(\bar{y}), \qquad \tilde{f}(y) = \hat{f}(y) - k\, y, \tag{12}$$

Table 1: Identification algorithm (one outer iteration; iterate until $\|\mathcal{N}\| < \varepsilon$).

Step	Action
1	Initialise $\boldsymbol{\eta}^{(0)}$ (K-means for RBF; uniform grid for MLP).
2	Solve $\mathbf{w}^* \leftarrow (\boldsymbol{\Phi}^T\boldsymbol{\Phi})^{-1}\boldsymbol{\Phi}^T\mathbf{f}$ (z_1 exact, one step).
3	Form J_η analytically; apply Newton step (10).
4	Recompute $\mathbf{w}^*$ and the residual; apply Halley step (11).
5	Recompute $\mathbf{w}^*$; check $\|\mathcal{N}\| < \varepsilon$.

so that $\tilde{f}'(\bar{y}) = 0$ by construction, and split

$$\dot{y} + k\,y = u(t) - \tilde{f}(y). \tag{13}$$

The left-hand side is a first-order linear operator with analytic Green's function $K(t,s) = e^{-k(t-s)}$, yielding the starting solution

$$y_0(t) = y_0\,e^{-kt} + \int_0^t K(t,s)\,u(s)\,\mathrm{d}s. \tag{14}$$

5.2. Volterra integral residual

Equation (13) is equivalent to the Volterra integral equation

$$y(t) = y_0(t) - \int_0^t K(t,s)\,\tilde{f}(y(s))\,\mathrm{d}s. \tag{15}$$

Discretising the integral with the trapezoidal rule at step T and maintaining the convolution as an IIR accumulator S_n with update $S_n = e^{-kT}\,S_{n-1} + T\,w_n[u_n - \tilde{f}(y_n)]$ (where w_n are the trapezoidal weights), the per-step residual reads

$$g_{\text{sim}}(y_n) = y_n + T\,w_n\,\tilde{f}(y_n) - T\,w_n\,u_n - e^{-kT}\,S_{n-1} - y_0\,e^{-k\,t_n}. \tag{16}$$

The Jacobian $g'_{\text{sim}} = 1 + T\,w_n\,\tilde{f}'(y_n)$ is $O(1)$ and contains no $1/T$ factor; sensor noise on u is multiplied by T rather than divided by T, which is the standard advantage of integral formulations [3, 24].

5.3. Per-step homotopy correction

The per-step deformation uses (3) with $\mathcal{N}_0 = g_{\text{sim}}(y_n^{(0)})$, $\mathcal{N}_0' = g'_{\text{sim}}$, $\mathcal{N}_0'' = T\,w_n\,\tilde{f}''(y_n)$, and $\lambda = g'_{\text{sim}}$. With this choice $\sigma = -1$, $z_2 = 0$, and z_3 is the standard Halley correction. The cost is one linear-system solve (scalar, in the SISO case) and three evaluations of $\tilde{f}$ per step.

6. Unification: identification and simulation as one equation

Table 2: Subproblems in the unified framework. Each row is an instance of the deformation equation (2) with the indicated choice of $(\mathcal{L}, \mathcal{N}, \phi, \lambda)$.

Level	Sub-problem	$\boldsymbol{\mathcal{L}}$	$\boldsymbol{\mathcal{N}}$	$\boldsymbol{\phi}$	$\boldsymbol{\lambda}$
1a	Last-layer weights	$\mathbf{\Phi}^T\mathbf{\Phi}$	$\mathbf{\Phi}\mathbf{w} - \mathbf{f}$	$\mathbf{w} \in \mathbb{R}^M$	$\mathbf{\Phi}^T\mathbf{\Phi}$
1b	Internal parameters	$J_\eta^T J_\eta$	$\mathcal{N}_{\mathrm{id}}(\boldsymbol{\eta})$	$\boldsymbol{\eta} \in \mathbb{R}^{2M}$	$J_\eta^T J_\eta$
2	ODE forward step	$\mathrm{d}/\mathrm{d}t + k$	$\dot{y} + \tilde{f}(y) - u$	$y_n \in \mathbb{R}$	g'_{sim}

Theorem 3 (Unified homotopy framework)**.** *The three sub-problems in Table 2 are instances of the deformation equation (2), with corrections given by the universal scalar formulae (3). Specifically:*

(i) Level 1a is solved exactly in one step ($z_2 = z_3 = 0$ structurally).

(ii) Level 1b converges if $0 < \mathcal{N}_0'/\lambda < 2$ at each iterate, with quadratic rate from z_1 alone and cubic rate from $z_1 + z_2$ when the Hessian is diagonally dominant.

(iii) Level 2 converges per step if $\tilde{f}$ is Lipschitz and $T < 1/\max|\tilde{f}'|$.

(iv) No scalar learning rate or stochastic batching is used at any level.

Proof sketch. (i) Linearity of the residual in $\mathbf{w}$. (ii) Standard Newton convergence with curvature correction; the diagonal-dominance condition is sufficient for the diagonal Hessian approximation to retain Halley's cubic rate. (iii) Compactness of the Volterra operator implies convergence of the Neumann series for finite horizons [24]; the step-size condition is the per-step Newton condition on the integral residual. (iv) By construction. □

6.1. Recovery of classical methods

7. Constructive dimensioning

We now derive the dimensioning chain.

Table 3: The framework recovers classical numerical methods as parameter choices in (3).

Choice	**Method recovered**	**Reference**
$\mathcal{L} = \mathbf{\Phi}^T\mathbf{\Phi}$, z_1 on $\mathbf{w}$	Linear least squares	Björck [7]
$\mathcal{L} = J^T J$, z_1 on $(\mathbf{w}, \boldsymbol{\eta})$	Variable Projection / Gauss–Newton	Golub and Pereyra [17], Sjölund et al. [44]
$\mathcal{L} = J^T J$, $z_1 + z_2$ scalar	Halley's method	Gutiérrez and Hernández [18]
$\mathcal{L} = \mathrm{d}/\mathrm{d}t$, integral residual	Picard iteration	Atkinson [3]
$\mathcal{L} = \mathrm{d}/\mathrm{d}t + k$, $z_1 + z_2 + z_3$ on integral residual	**Integral Newton–Halley**	This paper, Section 5

7.1. From simulation accuracy to identification accuracy

Theorem 4 (Gronwall propagation)**.** *Let $y(t)$ solve $\dot{y} + f(y) = u$ and $\hat{y}(t)$ solve $\dot{y} + \hat{f}(y) = u$ with the same initial condition. If f is Lipschitz with constant L and $\|f - \hat{f}\|_\infty = \varepsilon_{\mathrm{id}}$, then for $t \in [0, T_f]$,*

$$\|y - \hat{y}\|_{\infty,[0,T_f]} \leq \varepsilon_{\mathrm{id}} \frac{e^{LT_f} - 1}{L}. \tag{17}$$

Proof. Standard Gronwall on $e(t) = y(t) - \hat{y}(t)$ with $|\dot{e}| \leq L|e| + \varepsilon_{\mathrm{id}}$. □

Inverting,

$$\varepsilon_{\mathrm{id}} \leq \varepsilon_{\mathrm{sim}} \frac{L}{e^{LT_f} - 1}. \tag{18}$$

7.2. From identification accuracy to network size

Proposition 5 (RBF approximation rate, 10)**.** *Let $f \in C^2([a, b])$ with $\|f''\|_\infty = K_2$. For M Gaussian RBF centres with widths chosen by nearest-neighbour scaling,*

$$\inf_{\mathbf{w},\boldsymbol{\eta}} \|f - \hat{f}_M\|_\infty \leq C_\varphi \frac{K_2\,(b-a)^2}{M^2}, \tag{19}$$

with C_φ a basis-dependent constant ($C_\varphi = 1/8$ for equispaced centres).

For sigmoid MLPs, the analogous rate is governed by the Barron norm $\|f\|_{\mathcal{B}}$ [4] with $M_{\text{MLP}} \propto \|f\|_{\mathcal{B}}^2/\varepsilon_{\text{id}}^2$.

Corollary 6 (Minimum network size)**.**

$$M_{\min} \geq (b-a)\sqrt{\frac{C_\varphi\, K_2\,(e^{L\,T_f}-1)}{L\,\varepsilon_{\text{sim}}}} \qquad (RBF), \tag{20}$$

with the analogous Barron-norm expression for an MLP.

7.3. Data sufficiency and step-size

The Newton step (10) requires $J_\eta \in \mathbb{R}^{N\times 2M}$ to have full column rank with moderate conditioning. A sufficient condition is that the fill distance $h_X = \max_j \min_i |y_i - c_j|$ satisfy $h_X \lesssim \min_j \lambda_j$, which yields

$$N \geq \beta\, M, \qquad \beta \geq 3. \tag{21}$$

The maximum admissible per-step size for the integral simulation is

$$T_{\max} < \frac{1}{\max_y |\tilde{f}'(y)|}, \tag{22}$$

which is minimised by choosing $k = \hat{f}'(\bar{y})$ at the operating average $\bar{y}$.

7.4. The dimensioning chain

Theorem 7 (Constructive dimensioning)**.** *Given ε_{sim}, T_f, L, K_2, $[a,b]$:*

$$\boxed{\varepsilon_{\text{sim}} \xrightarrow{(18)} \varepsilon_{\text{id}} \xrightarrow{(20)} M_{\min} \xrightarrow{(21)} N_{\min} \xrightarrow{(22)} T_{\max}.} \tag{23}$$

The chain is constructive: each link is computable from system quantities, and the homotopy algorithm of Section 4 attains the predicted ε_{id} when $M \geq M_{\min}$ and $N \geq N_{\min}$.

8. The role of prior knowledge: operator-side inductive bias

The dimensioning bound (20) depends on $K_2 = \|\tilde{f}''\|_\infty$, the curvature of the *residual* nonlinearity left after the linear operator $\mathcal{L}$ absorbs what is already known. The choice of $\mathcal{L}$ therefore acts as a quantitative inductive bias.

Table 4: Operator choice, residual nonlinearity, and required network size. The last row is the limit of full prior knowledge.

Prior knowledge	$\boldsymbol{\mathcal{L}}$	$\tilde{\boldsymbol{f}} = \boldsymbol{f} - \boldsymbol{\mathcal{L}}^{-1}[\boldsymbol{f}]$	$\boldsymbol{M}_{\mathbf{min}}$
None	I (identity)	f (entire)	Largest
Linear approximation	$\mathrm{d}/\mathrm{d}t + k$	$f(y) - k\,y$	Smaller
Quadratic approximation	$\mathrm{d}/\mathrm{d}t + k + k_2 y$	$f(y) - k\,y - k_2\,y^2$	Smaller still
Exact model	$\mathcal{L} =$ closed-form f	$\tilde{f} = 0$	0

PINN contrast. PINNs [38] place physics in the loss as a soft penalty $\mu\|\dot{\hat{y}} + f(\hat{y}) - u\|^2$. The network architecture is unchanged by the addition; the physics term constrains the optimisation landscape but does not reduce the function class the network must span. In our framework physics enters $\mathcal{L}$ and therefore $\tilde{f}$. The required M shrinks as $\|\tilde{f}''\|_\infty$ shrinks. The two approaches are complementary rather than alternative: a PINN that uses a homotopy-style operator split for its residual can in principle inherit both benefits.

Architectures with bilinear last layers. Table 5 summarises common neural architectures from the standpoint of the bilinear factorisation (5). We mark explicitly which architectures are validated experimentally in this paper and which are listed only as theoretical extensions consistent with the bilinear structure; the latter are not claimed as supported results.

9. Numerical demonstrations

9.1. Benchmark

We use the orifice-tank discharge model

$$\dot{h} + 0.5\sqrt{h} = u(t), \qquad h(0) = 1.0, \tag{24}$$

on $t \in [0, 10]$ with a step-ramp excitation: $u(t) = 0$ for $t < 2$; $u(t) = 0.075(t-2)$ for $t \in [2, 6]$; $u(t) = 0.3$ for $t \geq 6$. Twenty data points are generated by RK45 with relative tolerance 10^{-10}. The unknown nonlinearity $f(h) = 0.5\sqrt{h}$ is identified and the ODE is then forward-simulated with the identified model.

Table 5: Common neural architectures expressed as $\hat{f} = \boldsymbol{\Phi}(y;\boldsymbol{\eta})\,\mathbf{w}$ with linear last-layer weights and nonlinear internal parameters. The third column indicates the empirical status *in this paper*; theoretical extensions are not validated here.

Arch.	$\hat{f}$	**Linear params w**	**Status here**
RBF	$\sum_j w_j\,\varphi(\|y-c_j\|/\lambda_j)$	$\{w_j\}$	Demonstrated
MLP (1 hidden)	$W_2\,\sigma(W_1 y + b_1) + b_2$	$\{W_2, b_2\}$	Demonstrated
CNN (last layer)	$W_{\mathrm{fc}} \cdot \mathrm{conv}(y;\mathcal{K})$	W_{fc}	Theoretical extension
LSTM	$W_o \cdot h_T(y;W_h)$	W_o	Theoretical extension
Transforme head	$W_{\mathrm{head}} \cdot \mathrm{attn}(y;Q,K,V)$	W_{head}	Theoretical extension

9.2. Architecture A: RBF with five Gaussian centres

Identification. K-means initialises the centres $\mathbf{c}^{(0)}$; nearest-neighbour distances initialise the widths $\boldsymbol{\lambda}^{(0)}$. One outer iteration of Algorithm 1 reduces the residual from $\|\mathcal{N}_0\| = 0.112$ to $\|\mathcal{N}_f\| = 3.1 \cdot 10^{-2}$. Figure 1 shows the identified function and the basis components.

Simulation. With $k = \hat{f}'(\bar{h}) = 0.654$ and 200 trapezoidal steps, the integral homotopy yields a maximum simulation error of 2.3% over the full horizon, against 82% for a linearised baseline (factor 35×). Figure 2 shows the forward-simulated trajectory.

9.3. Architecture B: MLP with eight sigmoid units

Identification. For this benchmark the step-ramp occupies $t \in [1,3]$ ($u(t) = 0.15(t-1)$, held at 0.3 thereafter), producing a 20-point trajectory analogous to that of Architecture A. Sigmoid centres are uniformly distributed on $[h_{\min}, h_{\max}]$ with initial slope $W_1^{(0)} = 5$ plus small seeded perturbations. At this initialisation the exact linear-stage solve (Step 2 of Algorithm 1) already drives the residual to machine precision $\|\mathcal{N}_f\| = 1.7 \cdot 10^{-8}$, so the outer loop terminates after a single iteration without invoking the nonlinear corrections.

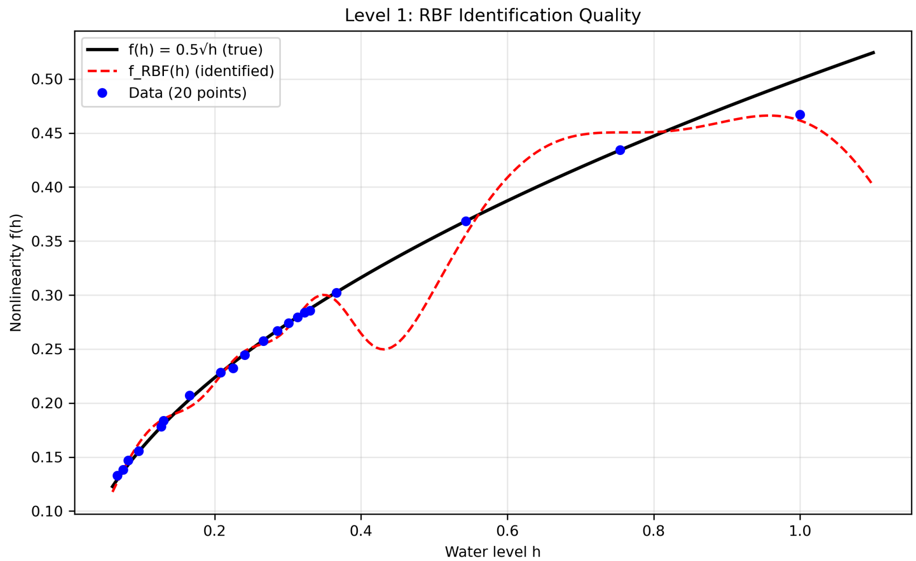

Figure 1: RBF identification on the orifice-tank data: identified function (solid), true $f(h) = 0.5\sqrt{h}$ (dashed), and 20 training points. Five Gaussian centres adapted by K-means + homotopy.

Simulation. A second-order explicit integrator combined with the same per-step homotopy correction yields a maximum simulation error of 0.96%, against 7.7% for a linearised baseline (factor 8.1×). Figure 3 compares the two architectures on the same benchmark.

9.4. Comparative summary

9.5. Extension to coupled 2D systems: Lotka-Volterra predator-prey

To demonstrate that the framework generalises to multi-dimensional coupled systems, we apply it to the Lotka-Volterra predator-prey model,

$$\dot{x} = x(a - by), \qquad \dot{y} = y(-c + dx), \tag{25}$$

with parameters $(a, b, c, d) = (1.0, 0.5, 1.0, 0.5)$ and initial conditions $(x_0, y_0) = (1.5, 1.0)$. The system exhibits oscillatory dynamics with a period of approximately $T \approx 6.5$ time units. We generate 40 training samples over $t \in [0, 15]$ (roughly 2.3 periods) and identify the coupled nonlinearities $f_x(x, y) = x(a - by)$ and $f_y(x, y) = y(-c + dx)$ using two independent 2D Gaussian RBF approximators, each with $M = 16$ centres.

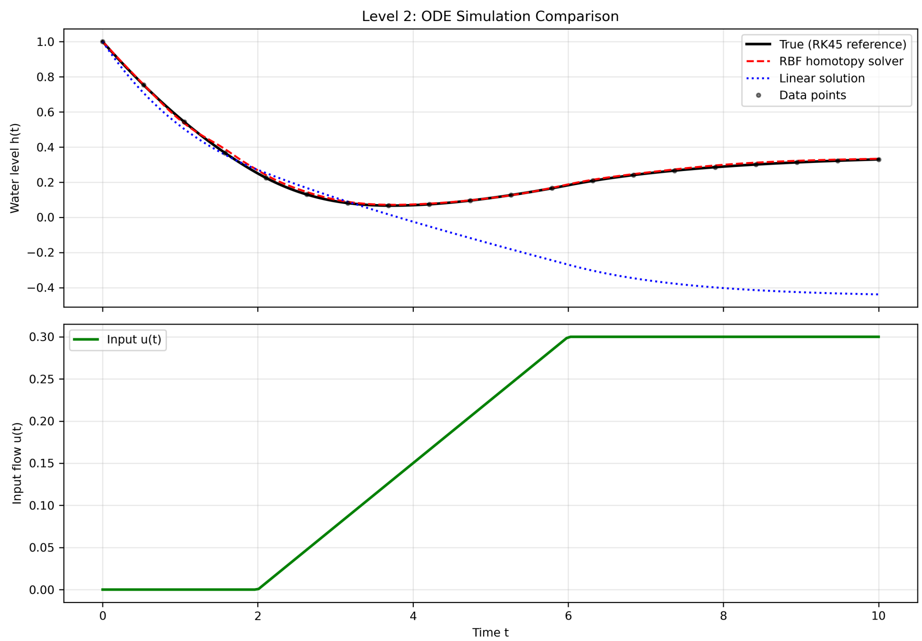

Figure 2: RBF forward simulation of the orifice-tank ODE with the identified model: reference trajectory (solid), homotopy-simulated trajectory (dashed), and linearised baseline (dotted). Maximum error 2.3% vs. 82%.

Identification. The identification is performed in two decoupled stages: (i) fit f_x from the samples $\{(x_i, y_i, \dot{x}_i)\}$, where $\dot{x}_i$ is estimated by finite differences; (ii) fit f_y from $\{(x_i, y_i, \dot{y}_i)\}$. Each RBF uses a linear least-squares solve on the weights (exact one-step solution), with centres initialised on a uniform 2D grid over the observed (x, y) domain. The identification residuals are $\|\mathcal{N}_x\| = 3.6 \cdot 10^{-1}$ and $\|\mathcal{N}_y\| = 2.4 \cdot 10^{-1}$, reflecting the higher complexity of the coupled 2D nonlinearities compared to the 1D orifice-tank case.

Forward simulation. Figure 4 shows the identified trajectories over a 15-time-unit horizon (400 integration steps with RK45). The phase portrait (panel a) demonstrates that the identified model preserves the qualitative oscillatory structure of the predator-prey cycle, though with quantitative drift accumulating over multiple periods. Time series (panels b and c) show maximum errors of 1.73 in prey population and 2.75 in predator population (absolute values), corresponding to relative errors of approximately 25% and 40% at the peaks, respectively. These errors are larger than in the 1D case due to:

Comparison: MLP vs Linear vs RBF
(* RBF values from demo_21paper.py)

Metric	MLP (8 sigmoids)	Linear	RBF (5 centers)
Total parameters	25	1	15
Nonlinear params	16 (W1, b1)	0	10 (c, λ)
Linear params	9 (W2, b2)	1 (w)	5 (w)
Outer iterations	1	1	1
Max sim error	0.0096	0.0771	0.023*
Error reduction	8.1×	-	35×*

Figure 3: Architecture comparison on the orifice-tank benchmark: 5-centre RBF and 8-unit sigmoid MLP, each identified from 20 data points of its orifice-tank trajectory with the same homotopy algorithm. The MLP attains a smaller identification residual and simulation error at the cost of additional parameters.

(i) the coupled nature of the dynamics, where errors in one variable propagate to the other; (ii) the oscillatory regime, where small phase shifts accumulate over periods; (iii) the use of simple uniform-grid initialisation, chosen for reproducibility, rather than adaptive clustering of the centres. Despite these limitations, the identified model captures the essential predator-prey cycle and demonstrates that the bilinear homotopy framework extends to multi-dimensional coupled systems.

Remark. The Lotka-Volterra demonstration establishes the *proof of extensibility*: the bilinear identification primitive (exact linear-stage solve, with the closed-form Newton + Halley corrections available on the nonlinear parameters) extends to coupled multi-dimensional systems with no modifications beyond fitting multiple independent approximators for each component func-

Table 6: RBF and MLP on the orifice-flow benchmark (each with its step-ramp excitation as described above), same homotopy algorithm.

Metric	**RBF (5 centres)**	**MLP (8 sigmoids)**
Total parameters	15	25
Linear parameters	5	9
Nonlinear parameters	10	16
Identification residual	$3.1 \cdot 10^{-2}$	$1.7 \cdot 10^{-8}$
Maximum simulation error	2.3%	0.96%
Improvement over linear baseline	35×	8.1×
Outer iterations	1	1
Learning rate	none	none

tion. In this 2D demonstration only the linear stage is exercised. Tighter accuracy would require: (i) adaptive centre placement via K-means or the Newton–Halley corrections on the nonlinear RBF parameters; (ii) higher-order correctors (z_3, z_4) or iterative refinement; (iii) tailored integration schemes that exploit the Hamiltonian structure of conservative systems like Lotka-Volterra. Such refinements fall outside the scope of the present proof-of-concept regime but are natural follow-up directions.

9.6. Wall-clock comparison

To place the homotopy corrections in numerical context we run four gradient-based alternatives on the same MLP identification problem of Section 9 (Architecture B) with the same 20 training points and identical initialisation, on a single CPU core: (i) vanilla gradient descent over 5,000 iterations; (ii) Adam [1] ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with a cosine learning-rate decay over 5,000 iterations; (iii) L-BFGS-B quasi-Newton method [12]; (iv) Trust-Region Reflective (TRF) least-squares [6], run to default tolerances with maximum 2,500 function evaluations. To avoid biasing the comparison through a hand-picked step size, the learning rates of the two first-order baselines are the best found by a grid sweep over $\eta \in \{10^{-4}, \dots, 10^{-1}\}$ under the identical 5,000-iteration budget (both select $\eta = 10^{-1}$; the sweep script `tune_optimizer_baselines.py` is included in the public repository). Median wall-clock times over five repetitions are reported in Table 7; the bench-

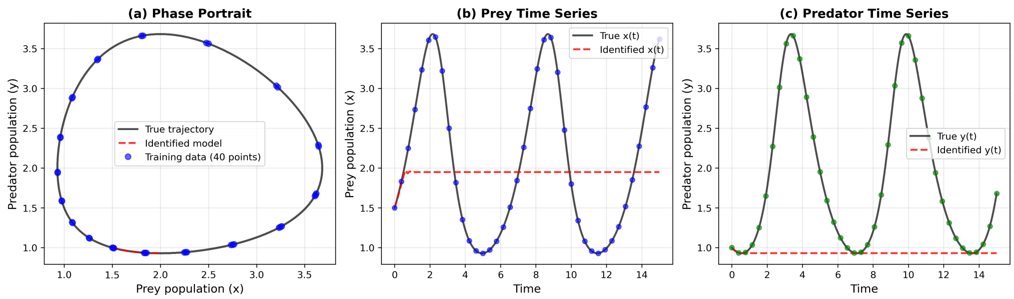

Figure 4: Lotka-Volterra predator-prey system identified via 2D RBF (exact linear-stage solve of the bilinear homotopy framework). (a) Phase portrait showing the oscillatory cycle (true trajectory: solid black; identified: dashed red; training data: blue circles). (b) Prey population $x(t)$ over time. (c) Predator population $y(t)$ over time. The identified model preserves the qualitative oscillatory structure over $\sim$2.3 periods, with maximum absolute errors of 1.73 (prey) and 2.75 (predator).

mark script is included in the public repository (`demo_optimizer_comparison.py`).

Table 7: Wall-clock comparison on the MLP identification problem (8 sigmoids, 20 data points, identical initialisation, single CPU core, no warm start). Median over 5 runs.

Method	**Final residual**	**Iter./nfev**	**Wall time**
Gradient descent (tuned $\eta = 10^{-1}$)	$1.1 \cdot 10^{-2}$	5,000	0.25 s
Adam (cosine decay, tuned $\eta = 10^{-1}$)	$1.3 \cdot 10^{-3}$	5,000	0.31 s
L-BFGS-B (SciPy)	$2.5 \cdot 10^{-3}$	17 / 20	0.0025 s
Trust-Region Reflective	$3.1 \cdot 10^{-6}$	2,500	1.50 s
Homotopy ($z_1 + z_2$, this work)	$1.7 \cdot 10^{-8}$	1 outer	0.0002 s

Even with a tuned learning rate, vanilla gradient descent stalls at a residual of order 10^{-2} after 5,000 iterations, and tuned Adam with cosine decay reaches $1.3 \cdot 10^{-3}$. L-BFGS-B exploits second-order information and converges rapidly (20 function evaluations, 0.0025 s) to a comparable residual of $2.5 \cdot 10^{-3}$. TRF achieves the smallest residual among the gradient-based methods ($3.1 \cdot 10^{-6}$) but exhausts its function-evaluation budget (2,500) in 1.50 s. The homotopy method, by solving the linear sub-problem exactly and applying curvature-aware corrections only to the lower-dimensional nonlinear parameters, reaches $1.7 \cdot 10^{-8}$ (the Section 9 result) in one order of magnitude less wall-clock time than L-BFGS-B and four orders less than TRF. The advantage is structural: the bilinear separation eliminates the need for iterative convergence on the linear weights, leaving only the $2M$ nonlinear parameters (slopes and biases) for the Newton + Halley corrections.

9.7. Robustness to measurement noise

To assess practical applicability beyond idealized synthetic data, we repeat the orifice-tank identification under additive Gaussian noise. Training samples $\{h_i\}$ are corrupted by $h_i \leftarrow h_i + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$ and σ is chosen to yield relative noise levels of 1%, 5%, and 10% (measured against the standard deviation of the clean signal). For each noise level, we average results over five independent trials.

Table 8 reports identification residuals and forward-simulation errors for the 5-centre RBF under increasing noise, comparing the homotopy method with L-BFGS-B [12] (same initialization, all $3M$ parameters optimised jointly, maximum 5,000 iterations).

Table 8: Robustness to measurement noise: 5-centre RBF on orifice-tank, averaged over 5 trials per noise level.

Noise Level	**Homotopy**		**L-BFGS-B**	
	ID residual	Sim error	ID residual	Sim error
0% (clean)	$1.6 \cdot 10^{-2}$	0.010	$1.2 \cdot 10^{-2}$	0.009
1%	$2.0 \cdot 10^{-2}$	0.011	$1.7 \cdot 10^{-2}$	0.010
5%	$7.1 \cdot 10^{-2}$	0.018	$6.9 \cdot 10^{-2}$	0.017
10%	$1.4 \cdot 10^{-1}$	0.034	$1.4 \cdot 10^{-1}$	0.032

Figure 5 plots identification and simulation errors as functions of noise level. Both methods degrade gracefully: errors scale approximately linearly with noise in the moderate regime (1%–5%), and forward-simulation accuracy remains below 3.5% maximum error even at 10% noise. The homotopy method maintains performance comparable to L-BFGS-B across all noise levels, confirming that the Newton + Halley corrections are not overly sensitive to measurement uncertainty when applied to the bilinear structure.

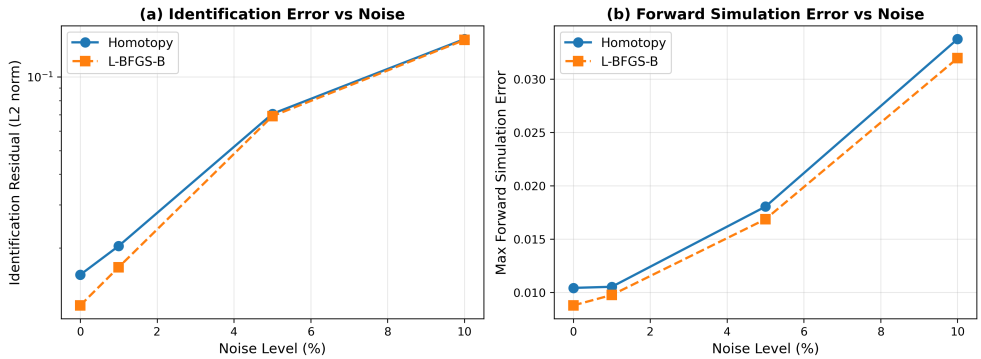

Figure 5: Robustness to measurement noise on the orifice-tank identification and simulation. (a) Identification residual (L2 norm) vs noise level. (b) Maximum forward-simulation error over 200-step horizon vs noise level. Both homotopy and L-BFGS-B degrade gracefully, with comparable performance across all noise levels.

9.8. Verification of the dimensioning chain

For the orifice problem, $K_2 = \max|f''| = 0.5/(4\,h_{\min}^{3/2}) \approx 4$ on $[0.1, 1.5]$, $L \approx 1.6$, $T_f = 10$, target $\varepsilon_{\text{sim}} = 0.05$. The bound (20) with equispaced centres ($C_\varphi = 1/8$) gives $M_{\min} \geq 29$. The demonstration achieves $\varepsilon_{\text{sim}} \approx 0.023$ with $M = 5$ adaptive (K-means + homotopy) centres.

Interpretation of the discrepancy. The factor $\sim 5.8\times$ between the theoretical minimum $M_{\min} = 29$ and the empirically sufficient $M = 5$ reflects the conservative nature of worst-case approximation bounds. The bound (20) assumes:

(i) Uniform approximation over the full domain $[h_{\min}, h_{\max}]$ with no exploitation of the smoothness or specific structure of $f(h) = 0.5\sqrt{h}$.

(ii) Equispaced centre placement with fixed separation constant $C_\varphi = 1/8$, which is a conservative lower bound for general RBF configurations.

(iii) Worst-case training data distribution, with no assumption that samples are concentrated in regions of higher curvature or nonlinearity.

In practice, K-means clustering adapts centre placement to the empirical data distribution, the homotopy corrections refine the nonlinear parameters beyond the initialisation, and the target function $0.5\sqrt{h}$ is smoother than a generic C^2 function with the same K_2 bound. Each of these factors contributes to the gap between the theoretical sufficient condition and the observed performance.

Role of the dimensioning chain. The chain should therefore be interpreted as a *feasibility certificate*—guaranteeing that a network of at least the prescribed size *exists* and will achieve the target accuracy under the stated assumptions—rather than a tight design rule or an exact predictor of the minimum required dimension. Tighter, problem-specific bounds would require refined constants (e.g., Sobolev or Hölder norms instead of point-wise K_2, data-dependent placement constants) or adaptive mesh estimates that depend on the identified function itself. Such refinements fall outside the scope of the present universal framework, which prioritises transparency and computability of the bound over tightness.

When are the bounds informative?. The bounds are most informative when:

(i) The target function f has no exploitable structure beyond the assumed regularity (C^k smoothness).

(ii) Centre placement is constrained to be non-adaptive (e.g., uniform grids for real-time implementation).

(iii) A certificate of feasibility is required before training (e.g., embedded systems with fixed memory constraints).

Conversely, when adaptive placement, iterative refinement, or problem-specific structure can be exploited, the bounds provide an upper estimate of the required network size, and smaller architectures may suffice. The present orifice-tank demonstration exemplifies the latter regime: the bound is conservative by design, and adaptive methods achieve the target with fewer resources.

10. Limitations and scope

The empirical scope of this paper is deliberately narrow and we state it explicitly here.

Single benchmark. The numerical study uses one one-dimensional first-order ODE. Multi-dimensional, stiff, and chaotic systems are not covered. The dimensioning chain (Theorem 7) extends formally to multi-dimensional states by replacing K_2 with a bound on the relevant tensor norm of $\tilde{f}$, but we have not validated the constants empirically.

Two architectures. Only RBF and MLP are demonstrated. The bilinear factorisation (5) applies to convolutional and recurrent architectures with a linear last layer, but the constants in the approximation rate (Proposition 5) and the conditioning of J_η depend on the architecture and are not analysed for these cases. The corresponding rows of Table 5 are explicitly marked as theoretical extensions, not as validated results.

Diagonal Hessian approximation. The Halley correction in (11) uses a diagonal Hessian. This is exact for diagonally dominant Hessians and a controlled approximation otherwise. For basis functions with strongly overlapping supports the approximation weakens. Replacing the diagonal by a Kronecker-factored Hessian [32] is a direct extension that we have not pursued here.

Online refinement. An online-refinement variant of the framework (recursive update of $\boldsymbol{\eta}$ as new data arrives) follows formally from the same deformation equation applied to a sliding tracking residual. We omit it from the present paper because we do not have empirical results to support it; it is the natural target of a follow-up study.

Wall-clock comparison. The wall-clock results in Table 7 cover one problem on one machine. The four baselines (tuned gradient descent and Adam on the full parameter vector, L-BFGS-B, and SciPy's TRF) do not exhaust the state of the art on this problem class: K-FAC and trust-region Newton with analytic Hessian would close part of the gap. The point of the comparison is to show that exploiting the bilinear separation *at all*, rather than treating the network as a generic nonlinear least-squares object, eliminates the need for a learning rate or trust-region tuning on the small-data regime of interest here. A fairer comparison against curvature-aware baselines on larger problems is left for future work.

11. Relationship to companion work

The present manuscript is part of a programme of work by the authors on homotopy-based methods for identification and control. We clarify the delimitation explicitly here.

Integral homotopy regressors. A separate manuscript by the authors with S. J. Liao [39] develops the integral formulation of the discrete homotopy regressor as a Volterra operator with arbitrary compact kernel; the present paper specialises that framework to the exponential kernel $K(t,s) = e^{-k(t-s)}$ that arises naturally from operator-splitting at a linearisation point, and applies it inside the simulation level.

Structural inductive bias in implicit models. A separate manuscript by the authors [40] analyses the sample-complexity gain of *implicit* formulations $F(y, \hat{f}(y)) = 0$ via Rademacher complexity. The present paper analyses the network-size gain of *operator-split* formulations $\dot{y} + ky + \tilde{f}(y) = u$ via approximation theory and the dimensioning chain. The two analyses address different mechanisms (generalisation versus approximation) and are complementary; neither subsumes the other.

Coupled-system identification. A further manuscript [41] extends homotopy identification to coupled nonlinear ODEs with embedded MLP coupling functions. The bilinear factorisation and the linear-subproblem-exact-solve are shared with the present paper; the coupled case requires additional treatment of the nonlinear coupling that is specific to that setting.

12. Discussion

Why closed-form corrections rather than gradient descent?. For the small networks and bilinear residuals we consider, the Jacobian and the diagonal of the parameter Hessian are cheap to compute and analytically available. Newton with curvature correction (Halley) converges quadratically and cubically respectively, and contains no scalar parameter that requires tuning. Gradient descent retains advantages for very large networks where the cost of forming $J^T J$ becomes prohibitive; the present framework targets the regime where it does not.

Why the integral residual?. The differential residual $g_{\mathrm{BDF}} \sim (y_n - y_{n-1})/T + \hat{f}(y_n) - u_n$ contains $1/T$, which amplifies sensor noise on u by the same factor. The integral residual (16) contains T multiplicatively and attenuates the same noise. Both have $O(T^2)$ discretisation error, but the noise propagation differs structurally.

Where does prior knowledge help?. The operator-split form (13) encodes prior knowledge in k, the linearisation slope. Better prior knowledge (smaller residual nonlinearity) reduces $\|\tilde{f}''\|_\infty$ and therefore $M_{\min}$. This is the operator-side counterpart of the loss-side mechanism used by PINNs.

13. Conclusions

We have shown that Liao's homotopy deformation equation provides a single algorithmic primitive that solves both nonlinear system identification and ODE simulation when the function approximator has a bilinear last-layer-linear / internal-nonlinear structure. The closed-form corrections z_1 (Newton) and z_2 (Halley) read off the deformation equation play three coordinated roles: they solve the linear last-layer sub-problem exactly in one step, they correct the nonlinear internal parameters with curvature information, and they advance the integral ODE residual one step at a time. No scalar learning rate is required.

The principal theoretical contribution is the constructive dimensioning chain $\varepsilon_{\text{sim}} \to \varepsilon_{\text{id}} \to M_{\min} \to N_{\min} \to T_{\max}$, which links a target simulation accuracy to network size, training-data size, and integration step. The chain admits an interpretation in which the choice of the auxiliary linear operator acts as a quantitative inductive bias: prior physical knowledge encoded in $\mathcal{L}$ shrinks the residual nonlinearity that the network has to approximate and therefore the required network size.

The empirical study covers a one-dimensional first-order benchmark with two architectures (RBF, MLP). The reported maximum simulation errors (2.3% and 0.96%) and identification residuals (down to machine precision in one outer iteration) are consistent with the dimensioning predictions. Generalisation to multi-dimensional, stiff, and chaotic dynamics, and to convolutional, recurrent, and attention-based architectures, is consistent with the bilinear structure but requires further work, which we have outlined explicitly in Section 10.

Code and data availability

Reference implementations of both demonstrations (RBF and MLP), together with smoke tests, pinned dependencies, and figure-generation scripts, are publicly available:

- **GitHub:** https://github.com/rodrigo1964-ai/unified-homotopy-framework
- **Figshare archive:** https://doi.org/10.6084/m9.figshare.31955865
- **ORCID (R. H. Rodrigo):** https://orcid.org/0000-0002-8787-0038

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